# Part A: Conceptual Understanding (Theory) — Answer Key

Machine Learning Fundamentals | Answers with Explanations & Diagrams

## Q1. What are Supervised Learning Algorithms?

Supervised Learning is a type of Machine Learning where the model is trained on a **labeled dataset** — meaning every input (feature) is paired with a known, correct output (target/label). The algorithm learns a mapping function from input to output by comparing its predictions to the actual answers and adjusting itself to reduce errors.

It is called "supervised" because the learning process is like a teacher supervising a student: the correct answers guide the model during training. Once trained, the model can predict outputs for new, unseen inputs.

**Examples:** Linear Regression, Logistic Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), K-Nearest Neighbors (KNN).
**Real-world use:** Predicting house prices (regression), spam email detection (classification), credit risk scoring.

## Q2. Difference between Regression Algorithms vs Classification Algorithms

Both are types of Supervised Learning, but they differ in the **nature of the output** they predict.

| Aspect | Regression | Classification |
|---|---|---|
| Output Type | Continuous numeric value | Discrete category / class label |
| Goal | Predict a quantity | Predict a category |
| Example Output | Price = Rs. 52,30,000 | Spam / Not Spam |
| Evaluation Metrics | MAE, MSE, RMSE, R-squared | Accuracy, Precision, Recall, F1-score |
| Common Algorithms | Linear Regression, Ridge, Lasso | Logistic Regression, Decision Tree, SVM |
| Use Case Example | Predicting temperature or sales | Detecting disease (yes/no), image labeling |

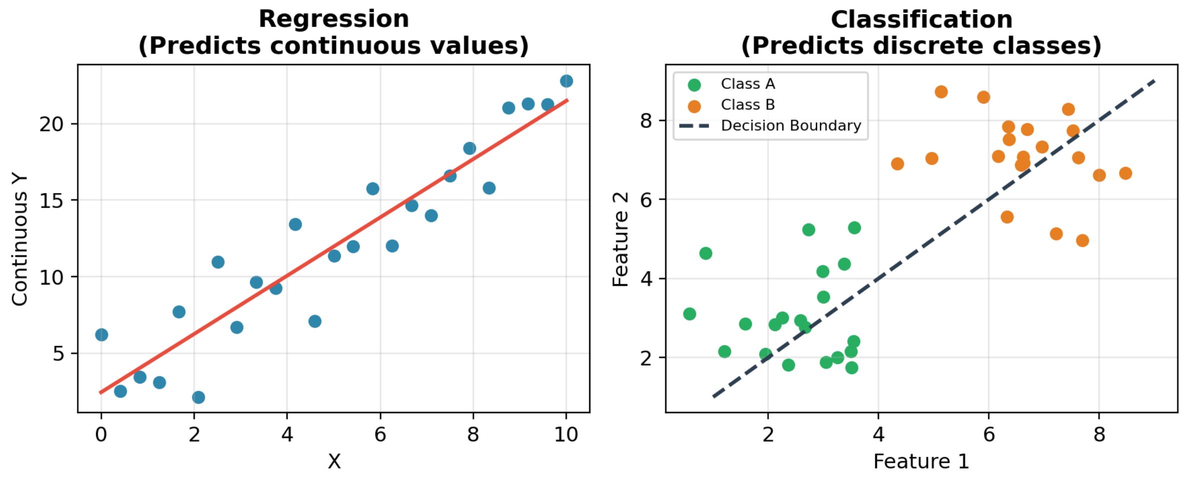


*Fig 1: Regression fits a continuous trend line; Classification separates data into distinct classes using a decision boundary.*

## Q3. Explain Simple Linear Regression

Simple Linear Regression is a supervised regression algorithm that models the relationship between **one independent variable (X)** and **one dependent variable (Y)** by fitting a straight line through the data points.

The relationship is expressed by the equation:

**Y = mX + c** (also written as Y = b0 + b1X)

where **Y** is the predicted output, **X** is the input feature, **m (or b1)** is the slope of the line (how much Y changes per unit change in X), and **c (or b0)** is the intercept (value of Y when X = 0). The best-fit line is found using the **Least Squares Method**, which minimizes the sum of squared differences between actual and predicted Y values.

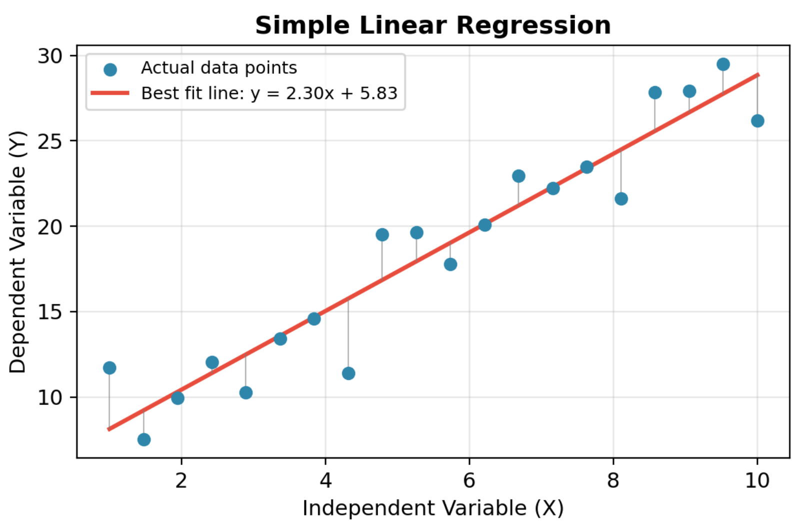


*Fig 2: The red line is the best fit; grey lines show the residuals (errors) that Least Squares minimizes.*

**Example:** Predicting a student's exam score (Y) based on the number of hours studied (X).

## Q4. List and Explain the Assumptions of Linear Regression

For Linear Regression to give reliable and valid results, the following assumptions about the data should hold true:

- **Linearity:** There must be a linear relationship between the independent variable(s) and the dependent variable — the change in Y is proportional to the change in X.
- **Independence of Errors:** The residuals (errors) should be independent of each other — the error for one observation should not influence another (no autocorrelation).
- **Homoscedasticity:** The variance of residuals should remain constant across all levels of X. If the spread of errors changes (funnel shape), it's called heteroscedasticity and violates this assumption.
- **Normality of Residuals:** The residuals should be approximately normally distributed, which is important for valid hypothesis testing and confidence intervals.
- **No Multicollinearity:** In multiple regression, independent variables should not be highly correlated with each other, as this makes it hard to isolate each variable's individual effect.
- **No Significant Outliers:** Extreme outliers can heavily distort the slope and intercept of the fitted line, leading to a misleading model.

## Q5. What is Bias–Variance Trade-Off?

The Bias–Variance Trade-off describes the balance between two sources of error that affect a model's ability to generalize to new data.

**Bias** is the error introduced by approximating a complex real-world problem with a simplified model. High bias means the model makes strong assumptions and is too simple, causing it to miss important patterns (underfitting).

**Variance** is the error introduced by the model's sensitivity to small fluctuations in the training data. High variance means the model is too complex and fits the noise in the training data rather than the true pattern (overfitting).

The **trade-off** is that decreasing bias (by making the model more complex) usually increases variance, and vice versa. The goal is to find the sweet spot of model complexity that minimizes **total error** = $Bias^2$ + Variance + Irreducible Error, giving the best generalization on unseen data.

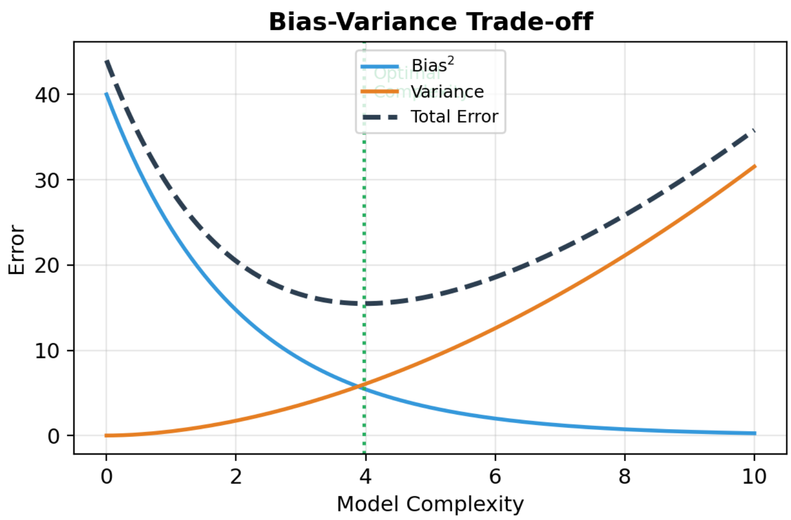


*Fig 3: As model complexity increases, bias falls but variance rises; total error is minimized at an optimal complexity level.*

## Q6. Explain Overfitting and Underfitting with Examples

**Overfitting** happens when a model learns the training data too well — including its noise and random fluctuations — resulting in a very complex model. It performs excellently on training data but poorly on new/test data because it fails to generalize.

*Example:* A very high-degree polynomial regression curve that wiggles through every single training point, or a decision tree grown so deep it memorizes each training example.

**Underfitting** happens when a model is too simple to capture the underlying pattern in the data. It performs poorly on both training and test data because it hasn't learned enough from the data.

*Example:* Fitting a straight line (linear model) to data that actually follows a curved, non-linear (e.g., sine-wave) pattern — the line misses most of the trend.

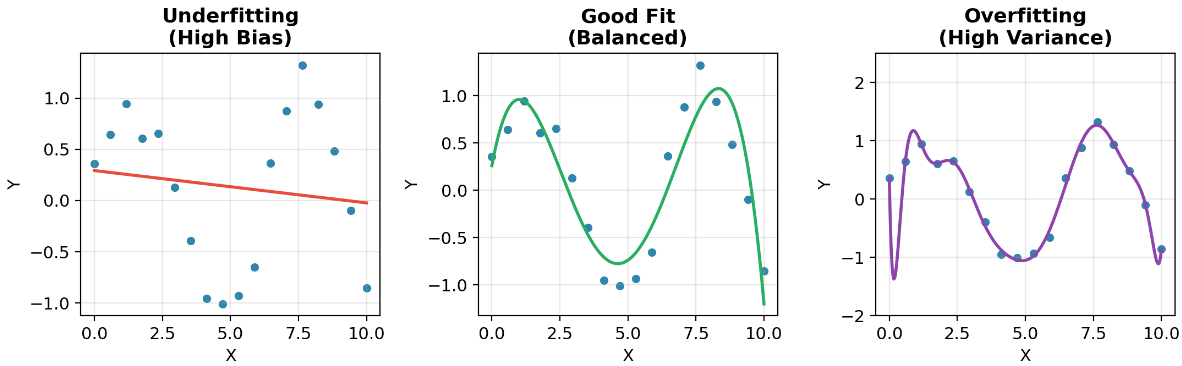


*Fig 4: Underfitting (too simple), a Good Fit (balanced), and Overfitting (too complex, fits noise) on the same dataset.*

**Solution:** The right balance — achieved through techniques like cross-validation, regularization (Ridge/Lasso), pruning, or choosing appropriate model complexity — helps avoid both extremes and results in good generalization, directly connecting back to the Bias-Variance Trade-off in Q5.